GlucoTwin

A synthetic patient twin for next-hour glucose alerts

Baseline EHR + CGM history + time-aware prediction

Anusha Gupta
IIT Kanpur
Team GlucoTwin | Solo proof of concept

Digital Twin Challenge 2026 | Phase 1 prototype

All patient records are synthetic. No established clinical validity.

Healthcare use case and prediction target

A doctor reviews patient context, recent glucose and a model-estimated alert at a chosen time

One operational question

When current glucose is below 180 mg/dL, will any reading reach that threshold during the next 60 minutes?

This threshold defines the prototype event. It is not a personalized treatment threshold.

Available at prediction time

- Baseline EHR recorded before the stream
- Two hours of CGM, including the current reading
- Current clock time
- No future readings or patient ID as predictors

Synthetic data and patient linkage

Project-generated records only; no external dataset, real patient records or live devices

Stream	Contents	Default scale
Baseline EHR	Demographics, diagnoses, BMI, HbA1c, duration	80 patients
Wearable CGM	Patient ID, timestamp, glucose (mg/dL)	161,280 readings

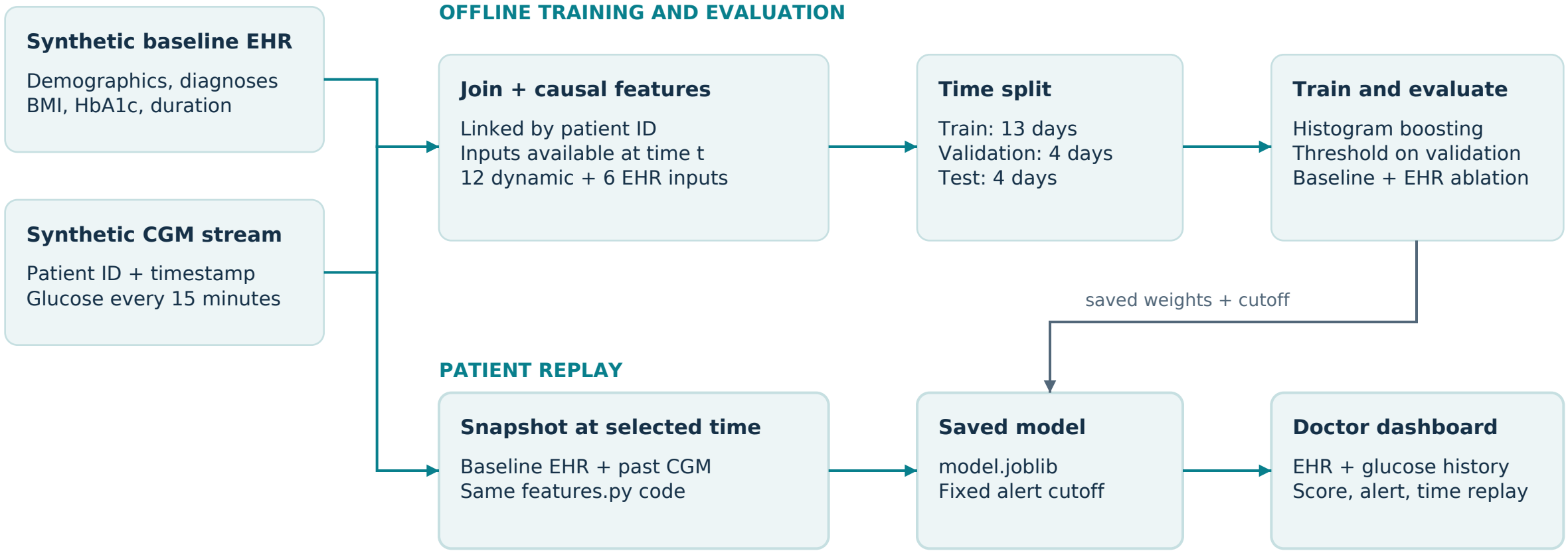
Fusion key: patient_id

Baseline labs are sampled before CGM. They are not calculated from future glucose.

The simulator includes meal-like rises, daily patterns and noise. Associations are engineered assumptions.

System architecture

Two linked streams feed the same causal feature code during training and patient replay



Future CGM creates labels only. It never enters prediction inputs; follow-up reveal is opt-in.

Synthetic demonstration only. Same patients across time splits; clinical validity is untested.

Chronological evaluation

Both history and future-label windows remain wholly inside their assigned time split

Split	Days	Eligible windows	Event windows
Train	1-13	47,339	1,657
Validation	14-17	14,209	514
Test	18-21	14,258	468

- Fit models on training data; choose alert cutoffs on validation F1.
- Freeze model settings and cutoffs before test evaluation.
- The same patients recur across splits; unseen-patient performance is untested.

Synthetic test results

Current-glucose baseline, wearable-only ablation and the preselected fused model

Model	ROC-AUC	Avg precision	Precision	Recall	F1
Current glucose	0.7584	0.1021	11.8%	23.1%	0.1557
CGM + clock	0.9537	0.4665	42.4%	54.5%	0.4771
CGM + clock + EHR	**0.9624**	**0.5023**	**45.6%**	**53.4%**	**0.4921**

Test event prevalence: 3.28%. Average precision makes the rare-event comparison visible.

Adding EHR changes simulator performance; this does not demonstrate clinical benefit.

Alert errors remain substantial

Fused model at its validation-selected cutoff; counts refer to overlapping test windows

Observed label	No alert	Alert
No event	TN 13,492	FP 298
Event	FN 218	TP 250

45.6%

Precision: event among alerted windows

53.4%

Recall: alerted among event windows

About half the event windows are missed at this operating point. A below-cutoff score does not mean safe.

Window-level metrics are not episode-level sensitivity, and repeated alerts have not been deduplicated.

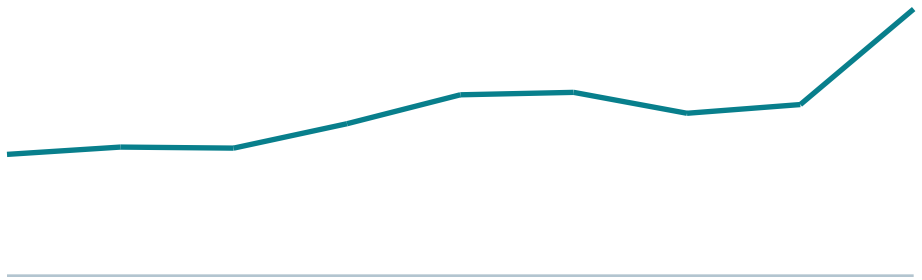
Doctor interaction with the twin

Workflow schematic using an actual validation snapshot; app.py contains the working interface

SYN_0001

Age: 57 years
BMI: 24.5
HbA1c: 7.1%
EHR: 2025-12-31
Diagnosis: Type 2 diabetes

Available CGM history



12:30 - 14:30 | mg/dL | no future readings

Model score

54.4%

Cutoff: 29.8%
Prototype alert for review

Snapshot: 15 Jan 2026, 14:30 | Synthetic local time

Select patient -> advance time -> inspect history and score -> optionally reveal observed follow-up

Revealed follow-up is never a forecast and never changes inference. Already-high states withhold new-onset scoring.

Limitations and next validation work

Implemented outcome: reproducible fusion, trained prediction and an updating doctor-facing patient state

Current limitations

- Synthetic patterns and engineered EHR associations
- Known patients across time splits
- Correlated windows; no episode-level evaluation
- No independent calibration or clinical validation

Future work

- Assess unseen patients and realistic missingness
- Study calibration and alert burden
- Validate externally under appropriate data governance
- Prospective evaluation before any clinical use

No real patient data, diagnosis, medication advice or regulatory approval is claimed.